

OWEN FORBES

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SUMMARY

Aerospace Engineering student with experience in CAD, MATLAB, and integrated mechanical-electrical systems. Interested in designing and testing high-impact aerospace systems.

EDUCATION

North Carolina State University, Raleigh, N.C.

2024-2028

- Bachelor of Science in Aerospace Engineering | GPA: 4.0/4.0

ENGINEERING INVOLVEMENT

SSailOR Project at NC State University | Undergraduate Researcher

2026-Present

- Designed and iterated mechanical components and assemblies in SolidWorks, incorporating structural and aerodynamic requirements while using XFLR5 aerodynamic analysis to evaluate design concepts.
- Fabricated carbon-fiber rods using composite manufacturing techniques, supporting the development of lightweight structural components.
- Supported experimental testing and data analysis using MATLAB and Python, using test results to evaluate and refine design changes.

American Institute of Aeronautics and Astronautics – Aircraft Workshop | Member

2025-Present

- Designed, built, and flight-tested radio-controlled aircraft from scratch.
- Applied aerodynamics, structures, and manufacturing principles in a hands-on team environment.

Vervara Network | 3D Modeler

2020-2022

- Designed and developed 50+ 3D models using Blockbench and Cubik Studio.
- Collaborated with a team of developers to refine and incorporate 3D models into interactive environments, ensuring optimal performance and visual accuracy.

PROJECTS

Sopwith Camel Radio-Controlled Aircraft | AIAA Aircraft Workshop

2026

- Developed a scaled R/C aircraft model of the Sopwith Camel using full-scale reference dimensions and imagery to establish accurate geometry and proportions
- Conducted flight testing and iterative design refinements to improve stability and control response.

Autonomous Mechanical Music Maker | Freshman Engineering Design Project

2025

- Designed and built an autonomous music-making system using a Raspberry Pi, solenoids, and an air pump to play a recorder automatically.
- Fabricated and assembled the aircraft with foamboard and wooden reinforcements, followed by flight testing and iterative refinements to improve stability and control response.

SKILLS

SolidWorks (CSWA) | Onshape | MATLAB | Raspberry Pi | CFD | XFLR5 | CAD Modeling | 3D Printing (Slicing) | Software | Mechanical Prototyping | Systems Integration | Team Collaboration